

Lou Chang

Backend-First Full Stack Engineer | Web Systems & Cloud Infrastructure

Toronto, ON, Canada

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Summary

Backend-first full stack engineer with 5+ years building and stabilizing web systems across healthcare, finance, B2B, retail, and legal workflows. Combines .NET/SQL and Python/FastAPI depth with Angular/Vue integration, Terraform, CI/CD, and AWS/Google Cloud delivery. Strongest at tracing problems across user workflows, application logic, databases, and infrastructure—cutting critical administration workflows from 5–8 minutes to 5 seconds and enabling local verification for a 40–50 API financial platform despite restricted client access.

Experience

Software Engineer — Backend & Cloud Infrastructure (Contract)

Independent Consulting

Toronto, ON
Sep. 2024 - Present

- Restructured a prototype FastAPI backend around domain, API, persistence, and provider boundaries; added SQLAlchemy repositories, Alembic migrations, Shopify onboarding, runtime validation, and quality gates.
- Refactored the Form 13.1 data-to-document flow, mapping DOCX/YAML metadata to FormKit fields and user input back to indexed DOCX fields; added dynamic rows, multiline handling, and behavior-focused tests for variable-length financial and property records.
- Rebuilt the Terraform deployment path from an AWS EC2/NGINX design to Google Cloud Run, Cloud SQL, Secret Manager, Workload Identity Federation, and Cloud Monitoring, with separated development and production resources.

Web Development Engineer

Duotify Inc.

Taipei, Taiwan
Jun. 2020 - Jan. 2024

- During annual conference registration, cut accounting, event, and membership administration workflows from 5–8 minutes to 5 seconds, fastest 0.4 seconds, on a Vue.js and ASP.NET MVC healthcare platform serving about 3,000 members by replacing multi-table AutoMapper/LINQ paths with SQL Server procedures, views, and functions.
- Designed 20–30 REST APIs for coupon, merchant, and membership workflows and aligned NSwag-documented contracts with an Angular 12 frontend, using JWT authentication, transaction-protected data changes, and Docker-based SQL Server development.
- Resolved reports of missing high-value orders on an iPad/Safari POS by tracing the issue to status-based filtering rather than data loss, correcting the query logic, and reducing repeat incidents.
- Built the company's first AWS development and test environment for a 40–50 API financial platform, enabling development and verification without depending on restricted client access before delivering 4–5 APIs and completing on-site acceptance.
- Mitigated a week-long traffic and scraper incident on a top-three Taiwan used-car marketplace with 50–70k monthly visits by using Azure Monitor to identify App Service saturation and configuring autoscaling during service recovery.

ERP Applications Engineer

Unitech Computer Co., Ltd.

Taipei, Taiwan
Mar. 2019 - May 2020

- Resolved cross-layer defects across AngularJS screens, ASP.NET MVC logic, and Oracle, SQL Server, and PostgreSQL data flows for a nationwide B2B platform covering product search, inventory, quotation, ordering, and member workflows.
- Built an Angular 8 and ASP.NET MVC 5 PoC with NG-ZORRO as a modernization reference for the legacy AngularJS platform, then documented and shared the Angular approach with the ERP engineering department.

Education & Certification

Seneca Polytechnic

Diploma, Computer Programming

Toronto, Canada
May 2024 - Aug. 2025

Hwa Hsia University of Technology

Bachelor's, Information Management

New Taipei City, Taiwan
Sep. 2018 - Jun. 2020

Microsoft

AZ-900: Azure Fundamentals

Taipei, Taiwan
Feb. 2022

Technologies

Backend, Data & Docs

Frontend & Client

Cloud & Delivery

Python, FastAPI, C#, .NET, Microsoft SQL Server, PostgreSQL, Oracle Database, SQLAlchemy, FormKit, DOCX XML
Angular, AngularJS, Vue.js, NG-ZORRO, REST APIs, OpenAPI / NSwag
AWS, Google Cloud, Cloud Run, Azure App Service, Azure Monitor, Terraform, GitHub Actions, Docker